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Marine Electrification - Barriers, Opportunities and Lessons Learned

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Electrification is reshaping short-sea shipping and mobility, driven by advances in technology and increased focus on reducing emissions and their negative impact on environmental and public health. With the proliferation of electric ferries, cargo ships, workboats, and tourism vessels, the industry is at a pivotal crossroads. This paper documents the evolving landscape of marine electrification, outlining achievements and lessons learned for key segments in Norway, the leading marine electrification market. Its objective is providing empirical, evidence-based guidance for transport authorities and operators seeking to leverage electrification for improved economic, environmental and societal performance of their marine fleets.

In the following we start by chronicling recent progress in electrification of key near-shore market segments like passenger and road ferries, workboats, tourism vessels, and cargo ships, providing case studies and techno-economic data as appropriate. We continue by identifying key technology, policy and commercial barriers and drivers that have come into play, extracting lessons learned for further proliferation of marine electrification. Next, we analyze the case for adoption of electrified marine transport in the Gulf Council Countries (GCC), considering climatic and geographical, economic and industrial characteristics. This includes techno-economic evaluation of electric vessels in key segments like crew transfer, waterborne mobility, short sea cargo, and fast workboat segments like pilot and patrol vessels. Finally, the paper outlines suggested policy measures based on lessons learned.

Introduction

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Figure 1—Sample electric vessels in operation in Norway.

Method

The paper relies on publicly available data, including academic publications and news stories, as well as interviews with industry, public and non-profit stakeholders internationally. Where conducive to the extraction of sensitive information, interviewees are granted anonymity. The insights of experts interviewed are supplemented by the author's own first-hand knowledge learned through his 20 years professional experience with transport electrification and marine technology. The collected data and observations will be subjected to qualitative and quantitative analysis, the results of which will be presented in the form of statistics, tables, and figures, as well as story lines for key domains.

Recent progress in Marine Electrification

Electrified marine transport has gained remarkable traction in markets where sustainability, policy leadership, and technological innovation intersect, with Norway standing as a global pioneer. More than 120 electric ferries, workboats, tourism boats and cargo vessels were deployed since 2015, generally performing their tasks without major issues. Below, an overview of recent developments.

Road ferries

The Norwegian Road Authority is responsible for 132 ferry connections across Norway, which are contracted out to private ferry operators through public tendering. By 2025, more than 40% of road ferry connections are electrified see figure [1], reducing annual CO₂ emissions from 300,000 tons in 2015 to a projected 149,000 tons in 2025, a reduction of 50% [2].

The transition started in 2015 with the deployment of the MF Ampere, the world's first fully electric road ferry, which marked a watershed moment. It provided practical experience and operational data benefitting subsequent projects on topics like route planning, battery system optimization, charging infrastructure, and grid integration. By 2025, it had executed more than 100,000 sailings.

Another example, Bastø Fosen operates one of the world's largest all-electric car ferries: Bastø Electric. 144 meter long, Bastø Electric shuttles up to 200 cars and 600 passengers across the Oslo fjord on Norway's most trafficked road ferry connection. It's 4.3 MWh battery is charged at each side of the connection using an automated charging connection providing up to 9 MW, enabling rapid turnaround. Along with its two all-electric sister ships, the operator saves 6 million liters of diesel per year. allowing Bastø Electric and its fleet to reliably transport over three million passengers and more than 1.5 million vehicles annually with minimal emissions. [3]

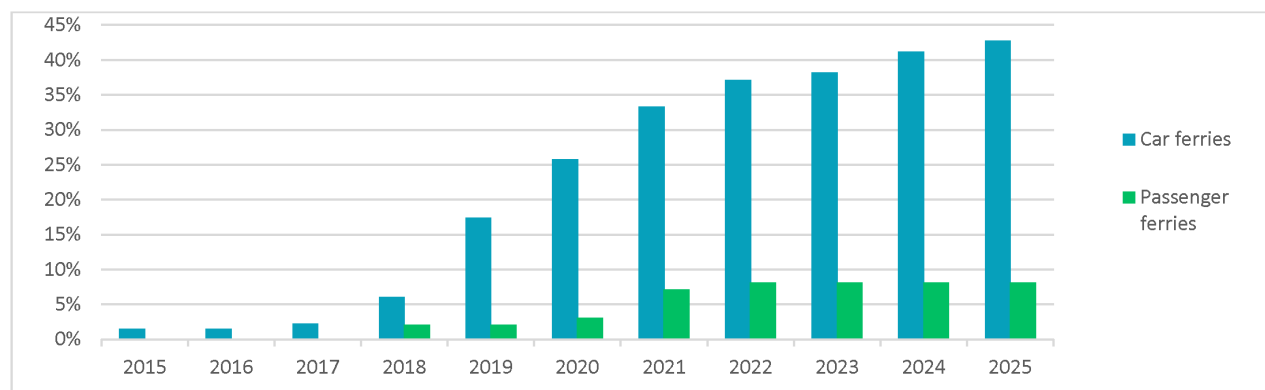


Figure 2—Norwegian electric ferries market share [1]

Launched in March 2023, MF Hydra is another innovative car ferry, measuring 82 meters and a beam of 17.5 meters. To meet the relatively long operating range required for the Hjelmeland–Skipavik–Nesvik on Norway’s west coast, MF Hydra’s combines battery power supply with a 400-kW hydrogen fuel cell generating power during its crossings. The vessel can rely solely on plug-in charging at the quay, solely on electricity generated by the fuel cells onboard, or a combination of the two. The vessel uses liquid rather than gaseous hydrogen fuel, as its higher density enables lower cost distribution and storage both onboard and on the quay. Its builder and operator had to go through a series of technology qualification steps to get the required approvals, world-first in commercial service, offering a technical template for future zero-emission [4].

Passenger ferries

Passenger ferries have also been electrified, starting out with urban routes. The city of Oslo already has 8 all-electric public transit ferries with a combined capacity of 2950 pax. The Oslo – Nesodden route is served by three electric ferries (pictured) with a combined capacity of 1800 pax in operation since 2019. The vessels feature battery 2MWh of battery energy storage each, and charge at up to 3MW using automatic connectors while docked. The next five vessels were commissioned in 2021 to service the city’s islands, each vessel powered by a 1MWh battery pack and automatic charging using pantographs while docked. The city of Fredrikstad has similarly commissioned 7 all-electric ferry boats.

The common denominator for these ferries, as well as the road ferries mentioned earlier, is that they operate at slow speeds, e.g. below 10 knots. Electrification has also arrived in the high-speed ferries and fast boats segments, which generally operate at 25 to 32 knots. Medstraum (Fig. 3), the world’s first fully electric fast ferry, began regular passenger service in Stavanger in 2023 cruising at 23 knots powered by two 550 kW motors energized by a battery pack with a gross capacity of 1524 kWh [4], which with a Depth of Discharge (DoD) of 80% provides a range of around 1 hour, e.g. 23 nautical miles.



Figure 3—MF Medstraum

The vessel's success demonstrates that electric propulsion is viable for certain high-frequency, short-range routes, provided that sufficient charging infrastructure is available and operational profiles are adapted to battery limitations.

Tourism vessels

Tourism vessels have also joined the electric wave, blending innovative design with environmental stewardship. The Vision of the Fjords, an award-winning tourism vessel built out of carbon fiber and propelled by a hybrid-electric propulsion system, has been navigating the UNESCO-listed fjords on Norway's west coast since 2016 with emissions dramatically lower than conventional craft [5], [6]. These projects have not only improved sustainability but also enhanced Norway's reputation as a leader in green tourism. Brim Explorer's silent, hybrid-electric ships glide through sensitive Arctic waters, offering guests immersive experiences with minimal disturbance to wildlife and spectacular landscapes [7].

Aquaculture

In the aquaculture sector, companies such as Moen Marine have championed the electrification of workboats, supporting sustainable fish farming operations along Norway's rugged coastline. After delivering its first electric fishing and aquaculture workboats in 2019, since 2022, the company's complete portfolio of aquaculture work vessels is offered with hybrid electric propulsion as standard [8]. The operating pattern of these vessels is a combination of cruising at higher speeds from shore to aquaculture installations at sea, where the vessels remain in place for extended periods of time. This makes these vessels well suited for hybrid electric propulsion, where diesel engines support the electric motors during cruise and batteries and electric motors take care of all maneuvering and hotel loads while working at site. These electric workboats have demonstrated reduced fuel consumption and noise, improving both environmental performance and crew comfort. Operators have documented maintenance reductions due to the simplicity of electric drivetrains compared to traditional diesel engines, although challenges remain in battery endurance on extended offshore assignments.

Short-sea logistics

Logistics and short-sea shipping are rapidly evolving as well, demonstrating that electrification is well suited for autonomous operations. Yara's Birkeland container vessel started operations in 2022. The 80 meter long, 15-meter-wide container vessel carries up to 120 TEU at a cruise speed of 6-7 knots using all-electric propulsion powered by 6.8 MWh of NMC batteries. The vessel is fitted with technology for autonomous navigation, mooring and charging, enabling fully autonomous operations monitored from a remote operation center in Horten, Norway [9]. Horten is also served by ASKO Maritime's two new autonomous, battery-powered roll-on/roll-off vessels for cross-fjord freight transport between the ASKO's logistics center on the east side of the Oslo Fjord and supermarkets on the west side. The 67-meter vessels can carry up to 16 Euro trailers and replace around 2 million truck kilometers per year. Propulsion is provided by a 500 kW electric motors, powered by 1846 kWh NMC batteries. As in the case of Yara Birkeland, the project leverages electrification and automation as key enablers for shifting cargo transport from road to sea, thereby reducing emissions and costs while improving road.

Barriers and Drivers

Despite impressive progress, the journey toward widespread marine electrification is marked by both barriers and drivers. This section discusses trends in enabling technology and innovation, regulatory frameworks and drivers, as well as the development of best practices within fleet electrification. It also examines economic considerations, including total cost of ownership, fuel savings, and maintenance implications. Through this exploration, we identify key obstacles and enablers that shape the adoption of electric vessels.

Enabling technology and innovation

Analyzing the Norwegian maritime electrification success, the fact that the required technology and solutions became available is fundamental. Although marine applications have additional requirements beyond those of automotive applications [10], the proliferation of electric cars, trucks and buses over the last decade and a half has undoubtedly played a key role in bringing to market the electric motors, power conversion and battery energy storage systems that paved way for marine electrification. The large numbers of electric ferries, work boats, cargo vessels and tourism vessels in daily operation clearly prove that electric propulsion batteries, power electronics and motors are maturing technologies. Moreover, lower fuel and maintenance costs and financial incentives means electric propulsion often delivers a favorable total cost of ownership, especially as energy markets evolve and carbon pricing mechanisms become more widespread.

However, not all marine electrification efforts have been successful. Norwegian designer and yard Brødrene Aa filed for bankruptcy in June 2025, after it had the contract for three 46-meter-high speed ferry newbuilds cancelled by the customer, Norled, due to failure to achieve the contracted 32 knots speed. To serve the 90 NM route from Trondheim to Kristiansund, the vessels were fitted with a novel battery swap solution developed by Norled, called Shift, with the goal of swapping batteries at the half-way stop at Brekstad. The solution incorporates four packs of NMC batteries with a total of 4 MWh, the battery packs alone weighed 28 tons [11]; combined with the required heavy duty steel support structure and the additional strengthening of the vessels' hulls and decks, we estimate that the total additional weight over a diesel powered vessel approached 50 tons. That's an estimated 50% increase in the lightship weight of a typical 46m carbon fiber catamaran.

To carry the added weight, the yard opted to extend the existing hull designs by 6 meters, but no new tank tests were carried out. The judgement was that since it had successfully deployed the same hull designs on multiple builds in the past, simplified extrapolation of previous design parameters sufficed [12]. However, sea trials showed the vessels topped out well below the target 32 knots of speed, and the contracts were cancelled.

This case study clearly shows that adding substantial extra weight of batteries must be matched with equally substantial moves to reduce the hydrodynamic resistance of the hull if the objective is to maintain or improve vessel performance characteristics, an observation also made more generally by the designers of the previously referenced Medstraum vessel [13]. However, since the Medstraum's relatively mild operational profile required just 1.5 MWh of batteries and 23 knots of cruise speed, it was sufficient to execute marginal improvements to the hulls' hydrodynamic resistance.

Key technology enabler 1: Reduced hull resistance. For longer range, higher speed routes, the Brødrene Aa experience shows more substantial improvements are required. Since 2017, the transport authorities of Nordland, Trøndelag and Vestland counties have run a competitive innovation program to deliver a vessel design with capacity of 150-200 pax and 40 knots service-speed that could achieve zero emission operation while requiring minimum 30% less power than the reference [14]. Entrants have proposed a variety of designs, including a fully foiling trimaran [15], a foil-assisted catamaran [16], and varieties of Surface Effect Ship hulls [17], [18]. Now in its fourth and final round of design optimization and verification, the objective of the counties is to realize newbuilds based on the new designs in 2026. Exceeding the objectives of the program, the realized vessels are expected to achieve up to 40% energy consumption compared to the reference, thereby enabling 40% increased range for the same battery capacity.

Key technology enabler 2: Increased battery energy density. Another key observation from the Brødrene Aa case is that higher energy density is critical for increased range and speed. The NMC batteries used added around 6 tons per MWh of capacity at the module level, which is what pushed Norled in the direction of battery swap. However, the additional weight of the battery casings and swap platform added another estimated 12-14 tons, meaning that the net weight addition was around 9 tons per MWh of battery. By contrast, semi-solid-state battery modules weighing just 4 tons per MWh have recently become available

[19]. Factoring in racks and other support and safety structures, this means the Brødrene Aa vessels would have achieved a 50% reduction in battery weight, and corresponding reduction in vessel drag and increase in achievable speed.

Combining the hull resistance and battery density improvements, it is now possible to achieve the 90 NM Trondheim-Kristiansund route without complicating execution with battery swap and at significantly higher speed than the targeted 32 knots.

Multi-stakeholder coordination and infrastructure execution

Since batteries are significantly less energy dense than diesel tanks, electric vessels have significantly lower range than diesel vessels. Therefore, electrification requires transport authorities and fleet operators to pay close attention to charging infrastructure. Based on the author's experience, Norway's experience in this regard can be considered somewhere between 70 and 80% successful. Key factors determining success or failure of marine infrastructure are outlined below.

Grid connection and charging infrastructure installation. Electric vessels typically carry between 0.5 and 10 MWh of batteries and should be charged while docked to ensure continued operation without disruption. This means power needs to be delivered to the jetty, the ease of which depends on the electricity grid. Successful projects are characterized by a combination of factors, such as deployment of battery back-up for the charging system (e.g. MF Ampera mentioned above), integrated project management by transport authorities and fleet operators as in the case of Oslo's urban ferries and accelerated permitting by port authorities and grid operators for interconnection. Negative experiences are characterized by unclear responsibilities, poor management and a failure to involve key stakeholders early in the process [20]. Delayed roll-out of infrastructure has in some cases been handled by executing all operations using back-up gensets, causing machine failures [21].

Technology maturity. Given the large batteries of electric vessels the power requirement is significantly higher than for electric cars and buses, which has implications for grid connection as mentioned above. It also means that apart from the smallest workboats, which can use the same chargers as cars, vessel charging technology has not yet been benefitting directly from the maturation of car charging infrastructure. The same is true for vessel-side charging communication technology; the relatively small size of the maritime market combined by more complicated certification processes has limited choices of components. This has led to significantly more expensive charging infrastructure than for road vehicles, causing some projects to experience significant delays [22].

The issues observed notwithstanding, it is the judgement of the author that successful executions far outnumber troubled ones with respect to charging infrastructure. This trend is supported by the increased standardization and integration between road and marine charging technology.

Regulations and policy

Ambitious decarbonization targets set by international bodies such as the International Maritime Organization [23], along with regional climate policies and emissions regulations, are compelling operators to accelerate the transition to electric fleets. Compliance not only avoids penalties but also enhances reputational value in a market increasingly sensitive to environmental credentials. Norway's achievements in marine electrification serves as a relevant case in point. Public policy and the push to reduce emissions have been fundamental in driving maritime electrification in Norway. The Road Authority has played an "entrepreneurial" role alongside the Norwegian Maritime Authority, Enova - the public fund for energy transition, and other national and regional public authorities "by moving beyond merely being a de-risker through playing an active role in market creation and transformation through public agencies and support schemes" [24].

Key policy driver 1: Emission reduction targets & incentives. In 2019, the Norwegian government set a goal to electrify all road ferries by 2025, later extending the deadline to 2030 [25]. These clear targets provide guidance to government and industry. As a result, investments in sustainable infrastructure have accelerated, and ferry operators are increasingly adopting new technologies to meet these ambitious environmental objectives.

Key policy driver 2: Public procurement. In addition to direct financial support and active market creation by public agencies, Norway has implemented a suite of targeted policies designed to lower the barriers to electrification and stimulate innovation across the maritime sector. For example, zero-emission requirements have been woven into public procurement contracts for ferry and express boat services, ensuring that newly commissioned vessels must meet rigorous environmental standards [24]. Competitive tendering frameworks have favored operators who can demonstrate not only compliance with emissions mandates but also plans for technological advancement and lifecycle sustainability. Furthermore, Norway's government has provided substantial grants for infrastructure development, including the rollout of high-capacity charging stations at key ports and the integration of shore power solutions. Incentives such as reduced port fees, exemptions from certain taxes, and prioritized access to routes have further accelerated the adoption of electric vessels. Collectively, these policies have created a robust ecosystem that rewards innovation, de-risks investment in new technologies, and positions Norway as a global leader in maritime decarbonization.

Marine Electrification in the GCC: Opportunities and Considerations

The Gulf Cooperation Council (GCC) region, comprising Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates (UAE), represents a promising frontier for marine electrification. Growing investments in renewable energy, transport electrification and other energy transition technologies, combined with the ascending global climate leadership of the UAE following its hosting of the Conference of the Parties (COP) 28 conference in 2023, make marine electrification increasingly relevant. Marine electrification offers a viable pathway to reduce emissions, improve air quality, and align with regional visions such as Saudi Arabia's Vision 2030 [26] and the UAE's Green Agenda – 2030 [27].

Opportunity 1: Passenger ferries

The region's economic landscape provides a compelling case for targeted electrification. For example, the routes of Dubai's public transit ferries are less than 12 NM in length, see Fig. 4.



Figure 4—Vision of the Fjords, launched in 2016 and operating using a hybrid-electric drivetrain is an award-winning pioneer in marine electrification in the tourism industry [5].



Figure 5—Norwegian electric ferries market share [1]



Figure 6—High density semi-solid state batteries. Source: Sealence.

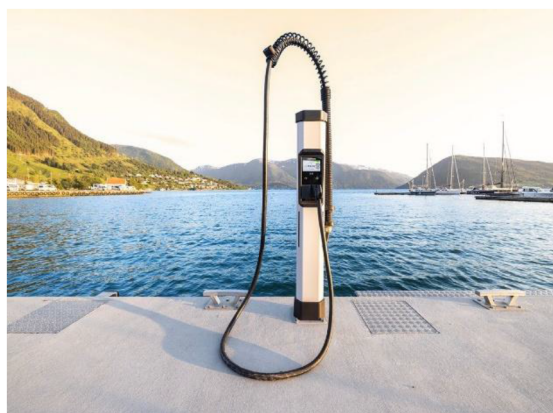


Figure 7—Charging infrastructure standardization between road and marine transport is well underway. Source: Kempower.

Similarly, Abu Dhabi is a city surrounded by water and is served by ferry services for passenger and vehicles between Saadiyat island and Al Aliah Island, and between Dalma island and Al Dhana port. It also operates water taxi services between popular destinations like Marsa Mina and Louvre Abu Dhabi, Yas Marina and Yas Waters Edge, Yas Bay and Yas Marina, and Yas Bay and Yas Waters Edge. Yet, as in the case of Dubai, none of these are yet electric. It is, however, clear that the distances and speeds required for these routes are highly realistic leveraging state-of-the art in technology and operational best practice.

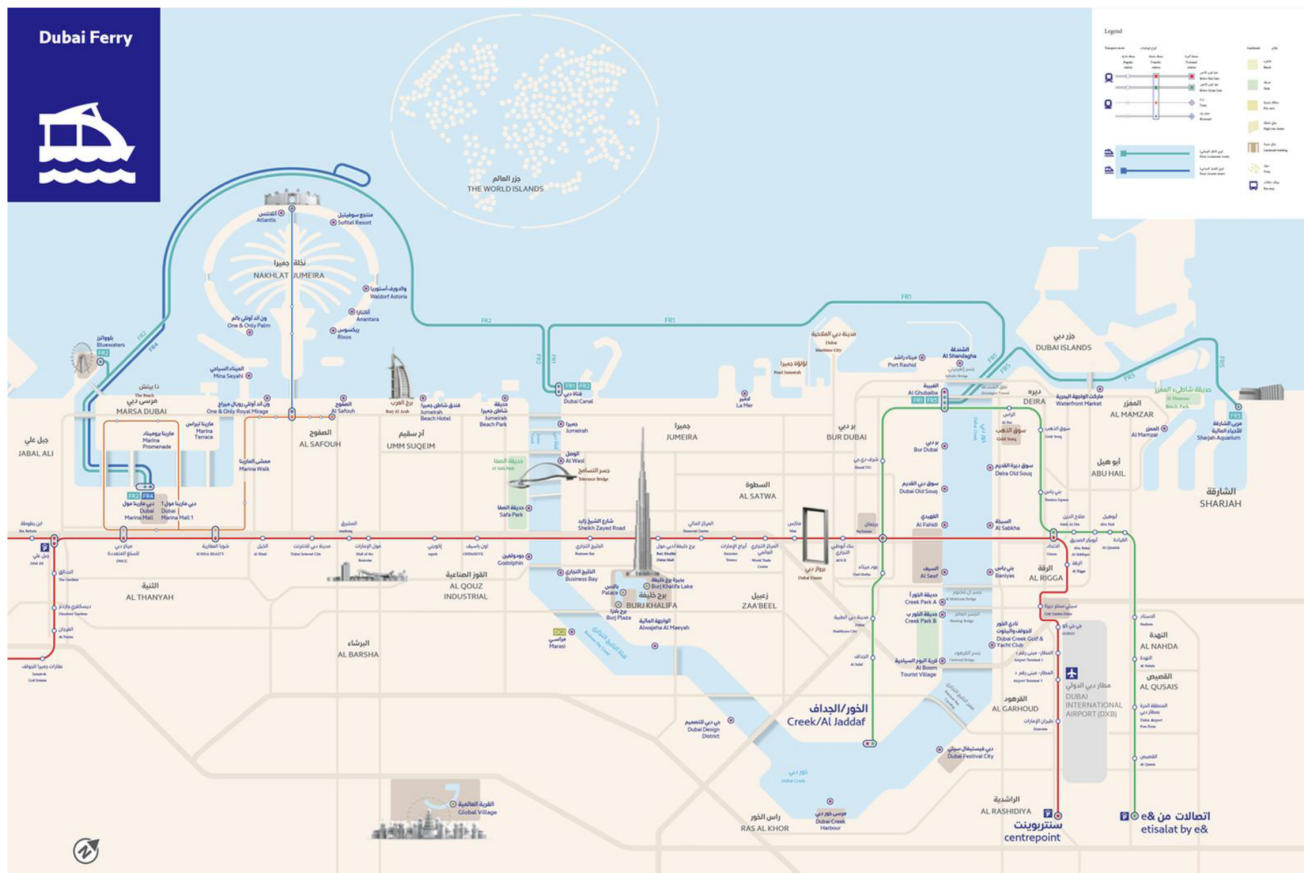


Figure 8—Dubai ferry routes [28].

Case study: High speed passenger ferry between Abu Dhabi and Dubai

Leveraging the state-of-the art in high-speed ferry technology, incorporating both hull resistance and battery density improvements, would enable creating a high-speed passenger ferry service between Abu Dhabi and Dubai. The distance between e.g. Saadiyat Island, home to popular destinations like the Louvre Abu Dhabi and Zayed National Museum, and the highly popular Dubai Marina area is around 55 nautical miles. At 40 knots, an all-electric speed ferry as described in the previous section would be able to cover the distance in around 1 hour and 20 minutes, consuming around 3000 kWh. Fitting the vessel with a net battery capacity of 6000 kWh would enable the following sample operating strategies:

- Charging at one end – the vessel would make a return trip without charging, then fully charging at 3-6 MW over 1-2 hours in Abu Dhabi upon returning.
- High power charging at both ends: Continuous service with high power charging at 6 MW for 20-30 minutes at both sides, e.g. during embarkation. 20-minute stops would enable the vessel to run two round trips for 6 hours and 40 minutes; 30-minute stops would enable non-stop operation, as the charging would completely replenish the energy consumed.

Opportunity 2: Crew transfer & workboats

Another key opportunity in the GCC is found in the workboats and crew transfer segments. All six GCC countries are involved in offshore oil and gas production, which invariably requires Crew Transfer Vessels (CTV) for carrying personnel between mainland and offshore installations and workboats for logistics, maintenance support, anchor handling & towing, emergency response and diving & ROV support operations. Given the relatively short distances between mainland and the offshore installations, these crew transfer and workboat operations seem ripe for electrification. They closely resemble the operations of

high-speed ferries and aquaculture service vessels, for which the Norwegian experience provides a clear electrification blueprint.

Case study: Pilot operations

The GCC is home to more than 25 significant seaports, and we estimate that the overall pilot vessel fleet size is between 150 and 200. These vessels operate within a port's approaches and anchorage zones, so their run distances are relatively short compared to open-sea voyages. Pilot launches are typically time sensitive, meaning these vessels typically operate at elevated speeds, typically above 30 knots. The table below provides an estimate of the distances these vessels must run; distances of 3 to 12 nautical miles are perfectly suitable for electrification.

Port	Pilotage Distance (NM)
Jebel Ali (UAE)	3–6
Ras Laffan (Qatar)	10–12
Hamad (Qatar)	11–12
Sohar (Oman)	6–8
King Abdulaziz (KSA)	8–10
Mina Salman (Bahrain)	3–5

However, pilotage vessels will often have relatively short time between jobs, and typically also serve other jobs like emergency response beyond pilotage, where high levels of autonomy are required. The combination of short routes with occasional need for very long range means a serial-hybrid setup with high power batteries providing the energy required for 70-80% of annual milage and generators providing the backup power when required. Leveraging the resistance-reducing hull technologies as mentioned earlier can reduce the battery capacity required for such a setup by 30-50%. Analysis of sample pilot routes using cost parameters relevant to the GCC context show the pilotage vessel operator can save 500-700 kUSD per year, including the cost of battery replacement caused by battery wear. That means the ROI for the added cost of electrification can be paid back in just 2-3 years (source: own analysis).

Region-specific challenges: climate

Adopting marine electrification in the GCC requires careful consideration of region-specific factors. The local climate, marked by high temperatures and humidity, means extra care must be taken to provide cooling for battery systems. Norway's extensive experience with electrification has delivered multiple lessons on how to provide inherently safe designs. Collaboration with international technology providers and class societies, such as DNV, are key to deploying these lessons in the GCC context.

Policy frameworks

The Norwegian experience shows electrification has been significantly driven by policy incentives and financial support. National and regional authorities in the GCC can learn from these experiences to facilitate marine fleet electrification. At the same time, marine electrification technology has significantly matured over the last decade, benefiting everybody that follow suit. Interviews with regional stakeholders have uncovered significant interest in adopting electrification technology but also laid bare the need for financial support to incentivize first movers and direct intervention to remove barriers to establishing the required charging infrastructure [29]. The latter could be executed by establishing demand-driven funds for operators seeking to deploy electric vessels; in return for financial support, authorities can require that the granted infrastructure is made available to all operators.

Conclusions

In conclusion, the advancement of high-speed, fully electric ferries represents a significant and practical strategy for achieving sustainable maritime transport within the Gulf Cooperation Council (GCC), especially on pivotal routes such as Abu Dhabi to Dubai. Through the integration of innovations in hull architecture, battery technology, and rapid charging infrastructure, it is technically viable to implement a zero-emission, high-frequency passenger vessel network that satisfies contemporary expectations for operational reliability and efficiency. The analysis indicates that, with a net battery capacity of 6000 kWh and access to high-power charging systems, these vessels could operate with minimal interruption, thus greatly mitigating greenhouse gas emissions relative to conventional marine modes.

Importantly, the experience of Norway offers valuable lessons for the GCC as it considers large-scale marine electrification. As a global leader in green shipping initiatives, Norway has demonstrated the effectiveness of coordinated policy support, public-private partnerships, and strong regulatory frameworks. Norway's comprehensive investments in charging infrastructure and continuous engagement with technology providers have also helped address operational challenges associated with harsh climates and demanding routes. These efforts have resulted in a robust and growing fleet of electric ferries, workboats, and utility vessels, providing a compelling example of how strategic investment and innovation can transform maritime transport systems.

Nevertheless, the distinct environmental and infrastructural circumstances of the GCC must guide the implementation process. Elevated ambient temperatures and humidity levels introduce considerable technical and safety challenges for large-scale battery deployments, necessitating robust thermal management systems and resilient energy storage solutions. Moreover, the expansion of grid capacity and provision of reliable energy supply at port terminals will demand coordinated investments and strategic planning among local authorities and regional energy stakeholders.

The findings of this work highlight the importance of international collaboration and knowledge transfer. Drawing from best practices and technological expertise from established leaders in electrified marine transport—particularly Norway—can substantially accelerate local adoption.

This study has incorporated a wide spectrum of source material and expert opinions, utilizing both qualitative and quantitative methodologies to articulate a comprehensive vision for the electrification of marine mobility in the GCC. As the international maritime industry intensifies its efforts toward decarbonization, it is imperative for the region to engage actively in research, demonstration projects, and capacity-building initiatives. Such engagement will be instrumental not only in meeting global climate objectives but also in shaping the trajectory of future maritime innovation.

The transition toward electrified marine transport should be viewed as both an economic and industrial opportunity, as well as an environmental necessity. By adopting a systemic approach that synthesizes technological advancement, infrastructure development, policy formation, and human capital enhancement, the GCC could emerge as a leader in sustainable shipping. The anticipated benefits extend beyond emission reductions to encompass improvements in air quality, energy security, and regional connectivity. The journey ahead will require strategic vision, sustained cooperation, and significant investment, but the prospective rewards for the maritime sector and broader society are substantial and enduring.

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